

Case Study — Fresh Graduate Job Portal

Resource Allocation & Risk Management Plan (PMBOK 8 Aligned)

Resource Allocation Plan

Project Overview

This project is a **job portal for fresh graduates** that provides features like **job search, resume upload, and basic AI recommendations**. Since the team consists of students, proper **resource planning** is required to manage time, workload, and efficiency.

Team Structure

- **Project Manager** → planning, coordination, tracking
- **Frontend Developer** → UI design and user interface
- **Backend Developer** → APIs, database, authentication
- **AI Developer** → job recommendation system
- **QA Tester** → testing and bug tracking
- **DevOps Engineer** → deployment and hosting

Project Phases & Resource Allocation

<i>Phase</i>	<i>Resources</i>	<i>Duration</i>
<i>Planning</i>	<i>Project Manager</i>	<i>1 week</i>
<i>Design</i>	<i>Frontend + Backend</i>	<i>2 weeks</i>
<i>Development</i>	<i>All Developers</i>	<i>6 weeks</i>
<i>Testing</i>	<i>QA + Developers</i>	<i>2 weeks</i>
<i>Deployment</i>	<i>DevOps</i>	<i>1 week</i>

Workload Distribution

- **Frontend Development** → 30%
- **Backend Development** → 35%

- **AI Module** → 15%
- **Testing** → 10%
- **Management** → 10%

Task Allocation

- **Frontend Developer** → build UI, integrate APIs, ensure responsiveness
- **Backend Developer** → develop APIs, manage database, implement security
- **AI Developer** → design and improve recommendation system
- **QA Tester** → perform testing and report bugs
- **DevOps Engineer** → deploy and monitor system
- **Project Manager** → manage tasks, deadlines, and communication

Resource Allocation Strategy

- Tasks are assigned based on **skills and expertise**
- Work is divided into **weekly sprints**
- Regular meetings ensure **progress tracking**
- **Buffer time** is added to handle delays

Resource-Related Risks

R6 — Skill Gap

Because some team members are new to certain technologies, there is a risk of **slow development**, which may delay the project.

Mitigation: **learning sessions and guidance**

R7 — Time Constraints

Because team members are students, there is a risk of **limited availability**, which may affect productivity.

Mitigation: **flexible scheduling and prioritization**

R8 — Single Dependency

Because some tasks depend on one individual, there is a risk of **work stoppage**, if that person is unavailable.

Mitigation: **knowledge sharing and documentation**

Tools Used

- **Jira** → task management
- **Slack** → communication
- **GitHub** → version control

Part 2 — Risk Management

Task 1 — Risk Identification (Cause → Risk → Effect)

R1 — Technical Risk

Because the team has limited experience with **AI systems**, there is a risk that job matching may be inaccurate, which could result in **poor user experience**.

R2 — External Risk

Because the system depends on **third-party APIs**, there is a risk of API failure or changes, which could result in **missing job data**.

R3 — Technical Risk

Because multiple users will use the system, there is a risk of **performance issues**, which could result in **slow response or crashes**.

R4 — Organisational Risk

Because team members have other commitments, there is a risk of **task delays**, which could result in **missing deadlines**.

R5 — Security Risk

Because sensitive data is stored, there is a risk of a **data breach**, which could result in **privacy issues**.

OPP1 — Opportunity

Because few platforms target fresh graduates, there is an opportunity to capture a **niche market**, leading to **high growth**.

Task 2 — Risk Analysis

ID	P	I	Score	Priority	EMV (\$)
R1	0.6	0.7	42	Critical	−3000
R2	0.4	0.8	32	High	−1600
R3	0.5	0.7	35	Critical	−2250
R4	0.5	0.6	30	High	−1500
R5	0.3	0.9	27	High	−1800
OPP1	0.5	0.8	40	Opportunity	+4000

EMV Summary

- Total Threat EMV = **−10,150**
- Opportunity EMV = **+4,000**
- **Net EMV = −6,150**

Recommended contingency reserve ≈ \$6,000

Task 3 — Risk Response Planning

R1 — AI Matching (Mitigate)

Use **rule-based logic** first, then improve AI with testing

- **Owner:** AI Developer
- **Trigger:** Accuracy < 70%
- **Budget:** \$500

R2 — API Dependency (Transfer + Mitigate)

Use **multiple APIs** and backup system

- **Owner:** Backend Developer
- **Trigger:** API failure > 48 hrs
- **Budget:** \$400

R3 — Performance (Mitigate)

Apply **load testing and caching**

- **Owner:** Backend / DevOps
- **Trigger:** Response time > 3 sec
- **Budget:** \$600

R4 — Delays (Mitigate)

Use **weekly tracking and clear tasks**

- **Owner:** Project Manager
- **Trigger:** missed deadlines
- **Budget:** \$300

R5 — Security (Mitigate)

Implement **encryption and authentication**

- **Owner:** Backend Developer
- **Trigger:** vulnerability detected
- **Budget:** \$700

OPP1 — Opportunity (Exploit)

Launch **early MVP** and promote to students

- **Owner:** PM + Marketing
- **Budget:** \$500